2023年卷 (期中)

1. (1) $\vec{n_1} = (1,2,-1)$, $\vec{n_2} = (1,1,0)$

$\Rightarrow \cos\theta = \dfrac{\vec{n_1}\cdot\vec{n_2}}{|\vec{n_1}|\cdot|\vec{n_2}|} = \dfrac{3}{\sqrt{6}\cdot\sqrt{2}} = \dfrac{\sqrt{3}}{2}$ 选D

$\Rightarrow \theta = \dfrac{\pi}{6}$.

(2). $V(t) = (\sqrt{t}, -\sqrt{2-t}, 1)$, $|V(t)| = \sqrt{3}$

$\Rightarrow L = \int_{\frac{1}{4}}^{\frac{1}{2}} |V(t)|\,dt = \dfrac{\sqrt{3}}{4}$ 选C

(3) 选C

$\lim\limits_{(x,y)\to(0,1)} \dfrac{\ln(1+xy)}{x} = \lim\limits_{(x,y)\to(0,1)} \dfrac{\ln(1+xy)}{xy}\cdot y = \lim\limits_{y\to 1} y = 1$.

$a=1$时 存 $f(0,1) = a = 1$. $\Rightarrow \lim\limits_{(x,y)\to(0,1)} = f(0,1)$

(4). $0 \le a_n \le \frac{1}{n}$. 选A

(A). $\sum\limits_{n=1}^{\infty} (-1)^n \dfrac{a_n}{(1+\ln n)^2}$: $0 \le \left|(-1)^n \dfrac{a_n}{(1+\ln)^2}\right| \le \dfrac{1}{n(1+\ln n)^2}$

存 $f(x) = \dfrac{1}{x(1+\ln x)^2}$ 为, 连续 $(x \ge 1)$

且 $\int_1^{\infty} \dfrac{1}{x(\ln(1+1))^2}\,dx = \int_1^{\infty} \dfrac{d(1+\ln x)}{(1+\ln x)^2}$

$= -\dfrac{1}{1+\ln x}\Big]_1^{\infty}$

$= 1$. 收.

$\Rightarrow \sum\limits_{n=1}^{\infty} \dfrac{1}{n(1+\ln n)^2}$ 收

$\Rightarrow \sum (-1)^n \dfrac{a_n}{(1+\ln n)^2}$ 绝对收敛, 故收敛.

(B). $\sum_{n=1}^{\infty}(-1)^n a_n$: 令 $a_n = \frac{(-1)^n+1}{2n}$. 则 $0 \le a_n \le \frac{1}{n}$

且 $\sum_{n=1}^{\infty}(-1)^n a_n = \sum_{n=1}^{\infty}\frac{1+(-1)^n}{2n}$

$= \sum_{n=1}^{\infty}\left(\frac{1}{2n} + \frac{(-1)^n}{2n}\right)$

发 发

$\Rightarrow$ 发.

(C) $a_n = \frac{1}{2n}$. $0 < a_n < \frac{1}{n}$. 但 $\sum a_n$ 发.

(D). $a_n = \frac{(-1)^n+1}{2n}$.

$\Rightarrow \sum (-1)^n \frac{a_n}{1+\ln n} = \sum \frac{(-1)^n+1}{2n(\ln n+1)}$

$= \sum \left(\frac{(-1)^n}{2n(\ln n+1)} + \frac{1}{2n(\ln n+1)}\right)$

对于 $\sum_{n=1}^{\infty}\frac{(-1)^n}{2n(\ln n+1)}$: $\frac{1}{2n(\ln n+1)}$ ↘. → 0. $\Rightarrow$ 收敛

对于 $\sum_{n=1}^{\infty}\frac{1}{2n(\ln n+1)}$: $\lim_{n\to\infty}\frac{\frac{1}{2n(\ln n+1)}}{\frac{1}{n\ln n}} = \frac{1}{2}$.

而 $\sum_{n=2}^{\infty}\frac{1}{n\ln n}$ 发 $\Rightarrow \sum_{n=2}^{\infty}\frac{1}{2n(\ln n+1)}$ 发

$\Rightarrow \sum_{n=1}^{\infty}\frac{1}{2n(\ln n+1)}$ 发.

(5). 若 $\sum_{n=1}^{\infty} a_n x^n$ 在 $x=-2$ 处发 则(C)

则 $\Rightarrow$ $|x|<2$ 时, $\sum_{n=1}^{\infty} a_n x^n$ 绝对收敛.

$\Rightarrow$ $|x|<2$ 时, $\left(\sum_{n=1}^{\infty} a_n x^n\right)'$ 绝对收敛.

$\sum_{n=1}^{\infty} n a_n x^{n-1}$

$\Rightarrow x=1$ 时, $\sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=1}^{\infty} n a_n$ 收敛.

② · (1). $r = \csc\theta\, e^{r\cos\theta}$

$\Rightarrow r\sin\theta = e^{r\cos\theta} \Rightarrow \boxed{y = e^x}$.

(2). $d = \dfrac{|1+2+3+4|}{\sqrt{1+2^2+3^2}} = \dfrac{10}{\sqrt{14}}$

(3). $v(t) = \int_0^t a(x)dx + v(0) = \left(e^x\Big]_0^t,\ \frac{x^2}{2}\Big]_0^t,\ -\frac{1}{2}\cos 2x\Big]_0^t\right)$ $+(1,0,0)$

$a(t) = (e^t, t, \sin 2t)$

$= (e^t - 1,\ \frac{t^2}{2},\ -\frac{1}{2}\cos 2t + \frac{1}{2}) + (1,0,0)$

$= (e^t,\ \frac{t^2}{2},\ \frac{1}{2} - \frac{1}{2}\cos 2t)$

$\Rightarrow r(t) = \int_0^t v(x)dx + r(0)$

$= \underline{(e^t - 1,\ \frac{t^3}{6},\ \frac{t}{2} - \frac{1}{4}\sin 2t)}$.

(4). $\sin x = x - \frac{x^3}{3!} + \cdots$

$\Rightarrow \lim_{x\to 0} \dfrac{3\sin(2x) - 2\sin(3x)}{6x^3 + x^4}$

$= \lim_{x\to 0} \dfrac{3\left(2x - \frac{(2x)^3}{3!} + o(x^3)\right) - 2\left(3x - \frac{(3x)^3}{3!} + o(x^3)\right)}{6x^3 + x^4}$

$= \lim_{x\to 0} \dfrac{(-4+9)x^3 + o(x^3)}{6x^3 + x^4} = \dfrac{5}{6}$.

(5). $S = \sum_{n=1}^{\infty} \dfrac{(-\ln 2)^n}{n!}$

$e^x = \sum_{n=0}^{\infty} \dfrac{x^n}{n!} \Rightarrow S = \sum_{n=0}^{\infty} \dfrac{(-\ln 2)^n}{n!} - 1$

$= e^{-\ln 2} - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$

3. $\sum_{n=1}^{\infty} \frac{\ln n}{n}(x-1)^n$.

(1). $\lim_{n\to\infty} \left| \frac{\ln(n+1)\cdot(x-1)^{n+1}}{n+1} \cdot \frac{n}{\ln n\cdot(x-1)^n} \right|$

$= \lim_{n\to\infty} \left| \frac{\ln(n+1)}{\ln n} \cdot \frac{n}{n+1} \right| \cdot |x-1| = |x-1| < 1$ 时, $\sum_{n=1}^{\infty} \frac{\ln n}{n}(x-1)^n$ 绝收

$x=0$ 时 $\Rightarrow \sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{n}$: $f(x)=\frac{\ln x}{x}$.

$f'(x)=\frac{1-\ln x}{x^2} < 0 \ (x\geq 3)$.

$\Rightarrow n\geq 3$, $\{\frac{\ln n}{n}\}\downarrow$, 且 $\lim_{n\to\infty} \frac{\ln n}{n} = 0$.

$\Rightarrow \sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{n}$ 收.

$x=2$ 时 $\Rightarrow \sum_{n=1}^{\infty} \frac{\ln n}{n}$: $f(x)=\frac{\ln x}{x} > 0$, $\downarrow (x\geq 3)$. 连续.

且 $\int_3^{\infty} \frac{\ln x}{x} dx = \frac{1}{2}(\ln x)^2 \Big|_3^{\infty} = \infty$, 发

$\Rightarrow \sum_{n=3}^{\infty} \frac{\ln n}{n}$ 发 $\Rightarrow \sum_{n=1}^{\infty} \frac{\ln n}{n}$ 发.

$\Rightarrow$ 收敛域: $0 \leq x < 2$. or $[0,2)$.

(2). $0<x<2$ 时, 绝对收敛.

而对于 $x=0$ 时, $\sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{n}$: $\sum_{n=1}^{\infty} \left| \frac{(-1)^n \ln n}{n} \right| = \sum_{n=3}^{\infty} \frac{\ln n}{n}$ 发.

$\Rightarrow x=0$ 时, 条件收敛.

4. (1). $\lim\limits_{(x,y)\to(0,0)} \dfrac{y^2\sin^2 x}{x^4+y^4}$

Sol. 不存在.

令 $y=kx(k\neq 0)$. 上式 $=\lim\limits_{x\to 0} \dfrac{k^2}{(1+k^4)}\cdot\dfrac{x^2\cdot\sin^2 x}{x^4}$

$=\lim\limits_{x\to 0}\dfrac{k^2}{1+k^4}\cdot\dfrac{\sin^2 x}{x^2}=\dfrac{k^2}{1+k}$.

(2). $\left|\dfrac{2xy}{\sqrt{x^2+2y^2}}\right|$ ## $=\left|\dfrac{x}{\sqrt{x^2+2y^2}}\cdot 2y\right|$

$\leq 2|y|$

$\Rightarrow (x,y)\to(0,0)$ 时, $2|y|\to 0 \Rightarrow \lim\limits_{(x,y)\to(0,0)}\dfrac{2xy}{\sqrt{x^2+2y^2}}=0$.

5. $L_1 = \begin{cases} x=t \\ y=1 \\ z=1+t. \end{cases}$ $L_2: \begin{cases} x=1+2s \\ y=0 \\ z=2s. \end{cases}$

$\Rightarrow \vec{T_1}=(1,0,1)$. $\vec{T_2}=(2,0,2)$

$\Rightarrow \vec{T_2}=2\vec{T_1} \Rightarrow L_1 \parallel L_2$.

$\Rightarrow$ 存在平面 M 过 L_1 与 L_2.

在 L_1 上取点 $A(0,1,1)$，在 L_2 上取一点 $B(1,0,0)$

$\Rightarrow \vec{AB}=(1,-1,-1)$.

$\vec{n_M}=\begin{vmatrix} i & j & k \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{vmatrix}=(1,2,-1)$，$M$ 过点 $B(1,0,0)$

$\Rightarrow$ 平面 M 为：$(x-1)+2y-z=0$

即 $x+2y-z=1$.

6. $0<b<a<1$

(1). $0<\dfrac{c_n}{\sqrt{(1+a^n)(1+b^n)}} \le a^n$

而 $0<a<1$，$\sum_{n=1}^{\infty} a^n$ 收 $\Rightarrow \sum_{n=1}^{\infty} \dfrac{c_n}{\sqrt{(1+a^n)(1+b^n)}}$ 收.

(2) $\left(\dfrac{a^n}{n^2}\right)^{\frac{1}{n}} < \left(\dfrac{a^n}{n^2}+\dfrac{b^n}{n}\right)^{\frac{1}{n}} \le \left(\dfrac{2a^n}{n}\right)^{\frac{1}{n}}$.

$\lim\limits_{n\to\infty}\left(\dfrac{a^n}{n^2}\right)^{\frac{1}{n}}=\lim\limits_{n\to\infty}\dfrac{a}{(n^{\frac{1}{n}})^2}$，$\lim\limits_{n\to\infty}\left(\dfrac{2a^n}{n}\right)^{\frac{1}{n}}=\lim\limits_{n\to\infty}\dfrac{2^{\frac{1}{n}}\cdot a}{n^{\frac{1}{n}}}=a$.

$=a$

$\Rightarrow \lim\limits_{n\to\infty}\left(\dfrac{a^n}{n^2}+\dfrac{b^n}{n}\right)^{\frac{1}{n}}=a$.

7. $r(t)=(\sqrt{2}\cos t,\sin t,\sin t)$.

Sol. (a) $v(t)=(-\sqrt{2}\sin t,\cos t,\cos t)$

$$|v(t)|=\sqrt{2\sin^2 t+\cos^2 t+\cos^2 t}=\sqrt{2}.$$

$$\Rightarrow T(t)=\frac{v}{|v|}=\frac{1}{\sqrt{2}}(-\sqrt{2}\sin t,\cos t,\cos t)$$

$$\Rightarrow \frac{dT}{dt}(t)=\frac{1}{\sqrt{2}}(-\sqrt{2}\cos t,-\sin t,-\sin t)$$

即 $\left|\frac{dT}{dt}\right|=\frac{1}{\sqrt{2}}\sqrt{2\cos^2 t+\sin^2 t+\sin^2 t}=1$.

$$\Rightarrow N(t)=\frac{dT}{dt}=\frac{1}{\sqrt{2}}(-\sqrt{2}\cos t,-\sin t,-\sin t).$$

(b). $k(t)=\frac{1}{|v(t)|}\cdot\left|\frac{dT}{dt}\right|=\frac{1}{\sqrt{2}}$.

8. $f(x)=\frac{4}{x^2-2x+5}+\ln x$

$$=\frac{4}{(x-1)^2+4}+\ln(1+(x-1))$$

$$=\frac{1}{1+\frac{(x-1)^2}{4}}+\ln[1+(x-1)]$$

$-1<x<3$

$\Updownarrow$

$|x-1|<2$

$\Updownarrow$

$\left|\frac{(x-1)^2}{4}\right|<1$.

即 $\frac{1}{1+\frac{(x-1)^2}{4}}=\sum_{n=0}^{\infty}\left[-\frac{(x-1)^2}{4}\right]^n=\sum_{n=0}^{\infty}\frac{(-1)^n}{4^n}\cdot(x-1)^{2n}$,

$\ln[1+(x-1)]=\sum_{n=1}^{\infty}\frac{(-1)^{n-1}(x-1)^n}{n}$, $-1\le x-1<1$ 即 $0\le x<2$

$\Rightarrow x\in(0,2)$ 时, $f(x)=\sum_{n=0}^{\infty}\frac{(-1)^n}{4^n}(x-1)^{2n}+\sum_{n=1}^{\infty}\frac{(-1)^{n-1}(x-1)^n}{n}$.